

STAT 461 Group Project Report

ANOVA of the Impact of Soccer Field Size on Goals Scored

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Executive Summary

This study investigates whether the size of a soccer field significantly influences the number of goals scored during a season. Using data from multiple professional leagues and categorizing fields into four size groups (small, medium, large, extra-large), we applied ANOVA to examine differences in average goals per game across these categories.

We first considered a blocked ANOVA design with league as a block, but we ended up selecting a simpler one-way ANOVA because of important assumption violations, especially the interaction between field size and league. We validated all necessary assumptions, including normality, homoscedasticity, and independence of residuals.

The results from our one-way ANOVA indicated no statistically significant effect of field size on goals scored, with a p-value of 0.55. This means we failed to reject the null hypothesis that average goals scored per game do not differ across field size categories. Additionally, effect size measures (e.g., $\eta^2 = 0.1196$) showed only a modest portion of variance in goals could be explained by field size.

Methodology

The purpose of this study is that we are attempting to answer the statistical research question: Does soccer field size have a statistically significant impact on the number of goals scored in a season? So, our null and alternative hypotheses would then be (where μ is the average goals per game for that field size category):

Null: Soccer field size has no statistically significant effect on goals scored.

$$H_0: \mu_{\text{small}} = \mu_{\text{medium}} = \mu_{\text{large}} = \mu_{x\text{-large}}$$

Alternative: Soccer field size does have a statistically significant effect on goals scored.

$$H_1: \text{There exist } i, j \text{ such that } \mu_i \neq \mu_j$$

We will also utilize an unusualness threshold of 0.05 and a type 1 error rate of 0.05 as is the general standard.

About Our Data

Observations about our data include that we have $k=4$ categories and subsequently $n-4 \rightarrow 20 - 4 = 16$ degrees of freedom. Our categorical factors are the stadium, team, size category, and league. Our numerical factors are the size in meters, games per season for a team, and goals for the 23-24 season. For our initial code to run some statistics, we will trim it down to just size category since L x W values will be hard to work with, as well as league, and goals columns. Our Hasse Diagram will only need two sources of variations in the middle and 4 boxes overall. Our exploratory data analysis will include the assumptions plots, generating the appropriate ANOVA table, and post-hoc analysis and tables if applicable. Our confirmatory data analysis will include analyzing all these things and making decisions for our hypothesis accordingly.

ANOVA Results

To perform our analysis of the impact of soccer field size on goals scored, we must first decide which particular ANOVA methods to utilize. Well, we have a response variable being the average number of goals scored per game in the season, a categorical factor of interest being field size, and an attribute to monitor being the league that the games are played in. So, our options are to either perform ANOVA + Block, except that it would be an incomplete block design, or we could perform One or Two-way ANOVA and accept the potential confounding that may happen between league and goals scored. Although it seems like the variable league is a covariate and ANCOVA may be useful here, ANCOVA requires a quantitative covariate, not a categorical one.

Because there seems to be a clear relationship between league type and field size, as seen in Figure 1, we can try to proceed with ANOVA + Block where the block is the league type.

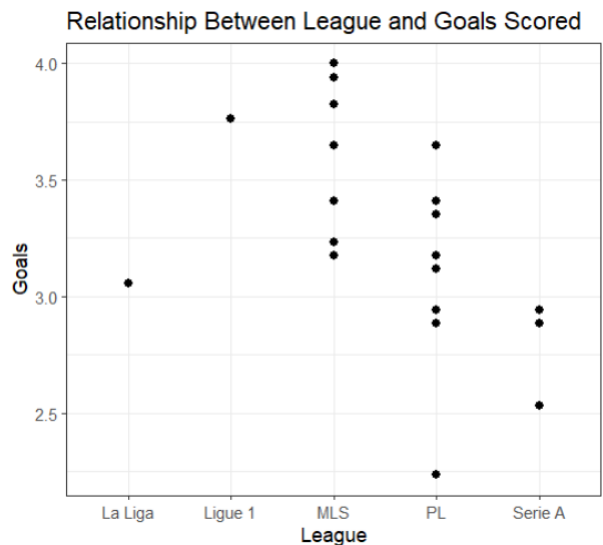


Figure 1 - Strip Chart of League and Goals Scored per Game

This may not be the most powerful test possible, but we feel that it can help us answer the SRQ most effectively while also accounting for notable attributes within the data. Thus, we can analyze the effects of our factor of interest while blocking the effects that league has on the number of goals scored. So, we can go through the assumptions for this block design to see if it is applicable. If it's not, then we can proceed with typical One or Two-Way ANOVA.

Let's begin by assessing the six ANOVA + Block requirements (would be seven for complete block design). Firstly, we must have a qualitative factor and quantitative response, which is satisfied since our response (goals scored per game) is quantitative, and our factor (field size) has been changed to categorical. Next, we must have an additive model along with estimable effects and estimable errors/residuals, which can be validated using a Hasse Diagram. As we can see in Figure 2, we can indeed create a Hasse Diagram from this model, and there are positive degrees of freedom on all nodes.

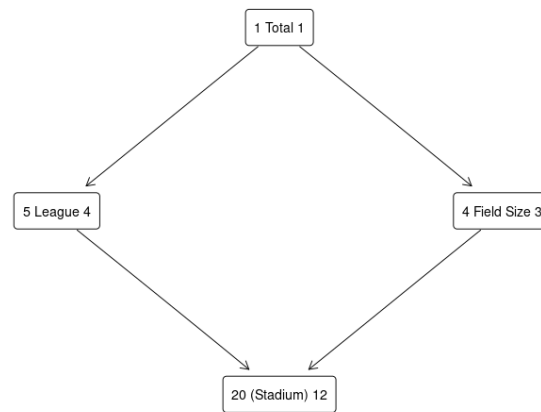


Figure 2 - Hasse Diagram

And the final ANOVA + Block requirement is that we have a categorical attribute (block) that creates sets of units that are similar to each other in some way. The league attribute creates sets of fields that are played on by players of similar pedigree and skill level, so this requirement is satisfied.

Now that we have satisfied the basic requirements that are necessary for ANOVA + Block, we must also assess some assumptions before proceeding with our modeling. The first assumption that we should test is the interaction between the block and the factor, which can be tested using the interaction plot in Figure 3.

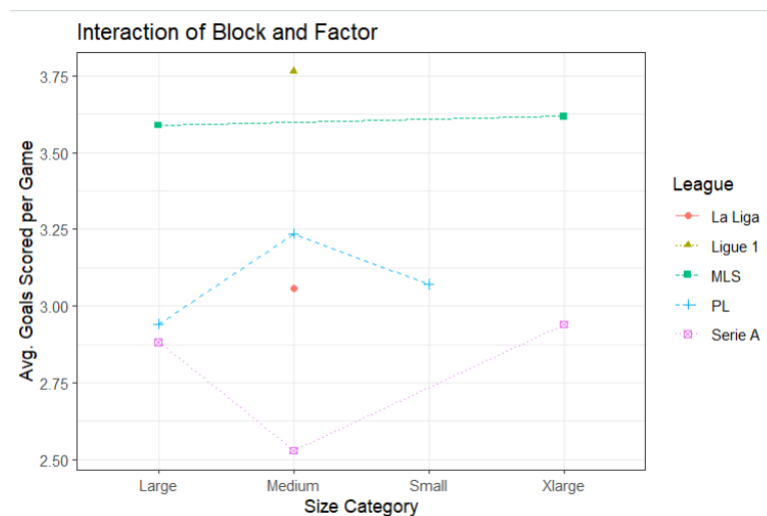


Figure 3 - Interaction Plot

As we can see, this plot doesn't look very good since we are looking for similar trends in slopes between the different leagues. So, it appears that ANOVA + Block may not be the best design for this circumstance. Luckily, we can still transition to a One or Two-Way ANOVA fairly easily from here since we haven't assessed any other assumptions. Because there is such imbalance between the leagues (e.g. only one size category for Ligue 1), the best option at this point appears to be to just drop the league attribute and perform One-way ANOVA. In an ideal world, our data would contain a lot more entries and leagues with different categories for each, and we would be able to perform a complete randomized block design, but that is not applicable with our current situation. So, let's do simple One-way ANOVA.

Proceeding with One-way ANOVA, the first base ANOVA assumption is Gaussian residuals, which we can evaluate using the QQ plot below in Figure 4.

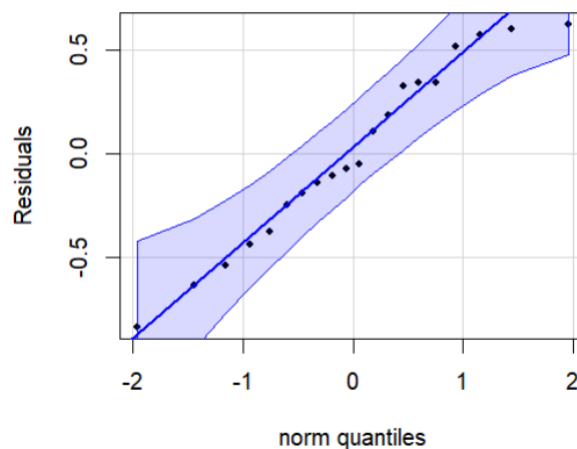


Figure 4 - QQ Plot

Since none of the points are outside of the 90% confidence envelope, we can say that this assumption is satisfied. The next assumption is homoscedasticity, which can be validated with a strip chart like the one in Figure 5.

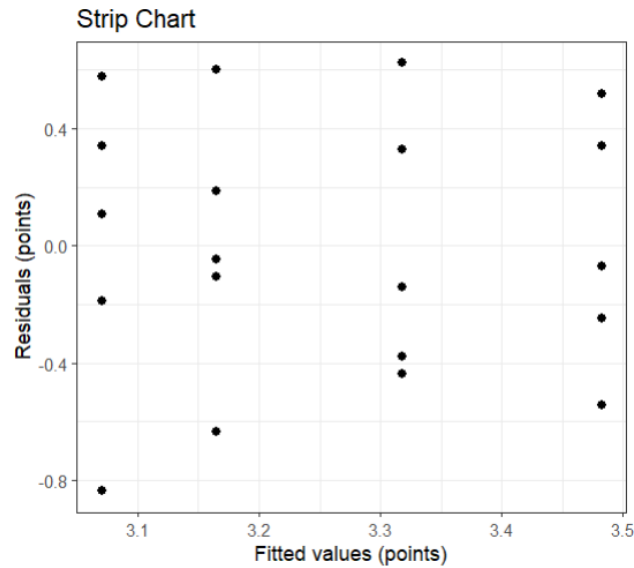


Figure 5 – Strip Chart

In the plot, we can see that there are no funnel or megaphone patterns in the plot and all categories have similar vertical space in the chart. Thus, homoscedasticity is satisfied.

The final base ANOVA assumption that needs satisfaction is independence of observations, which we can check with the index plot in Figure 6.

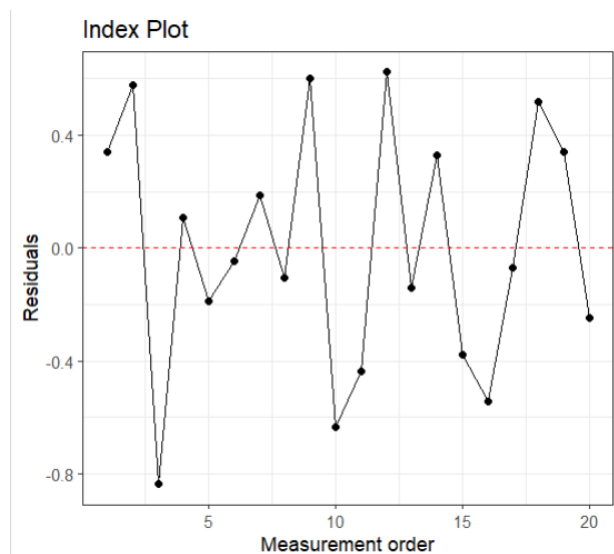


Figure 6 - Index Plot

Since there does not appear to be any clear patterns in the index plot and we got a Durbin-Watson statistic value of about 2.4, we can say that this assumption is satisfied.

Now that we have satisfied all of the necessary assumptions, we can begin our actual analysis. Our One-way ANVOA model with the field size category as the factor and the number of goals scored that season as the response results are in the ANOVA table below in Figure 6. We utilized the parametric shortcut ANOVA F-Test to answer our research question at a 5% significance level. Do note that since we are just doing One-way ANOVA, Type I, II, and III sums of squares give the same result.

Source	SS	df	MS	F	p-value	Eta Sq.	Omega Sq.	Epsilon Sq.
Size.Category	0.4886	3	0.1629	0.7244	0.5521	0.1196	0	0
Residuals	3.5972	16	0.2248	NA	NA	NA	NA	NA

Figure 7 - Advanced ANOVA Table

As we can see from the F-Statistic value of approximately 0.72 in the ANOVA table above, field size accounts for about 72% as much variation in goals scored as the residuals do. And when we see the p-value of 0.55, it becomes clear that we should fail to reject the null hypothesis that there is no statistically significant effect of field size on goals scored that season. This p-value of 0.55 means that under the null hypothesis, we would observe a test statistic value higher than or equal to 0.72 about 55% of the time. Since this is far greater than our unusualness threshold of 0.05, we should fail to reject the null hypothesis.

If we examine the effect size measures in the table, the Eta squared metric is 0.1196, which can be interpreted as: about 11.96% of the variance in the response variable average goals scored is explained by the factor field size. This, and the fact that our other effect size measurements are 0 (or at least close to 0), shows that the explanatory power of our factor (field size) is minimal. Thus, our non-significant p-value and effect size values lead us to conclude that field size category does not meaningfully explain variation in average goals scored per match for this particular dataset.

Normally, at this point, we would continue our analysis with pairwise or contrast Post Hoc analysis. But since there were no significant findings in our parametric shortcut F-Test, we have no evidence to follow up for our analysis. If we were to do Post Hoc analysis from here, it would

likely increase the Type 1 error rate and would thus be counterproductive for our overall analysis.

Discussion

While not a very big limitation, our data size was on the smaller end with 20 stadiums sampled with 16 degrees of freedom for within groups or residuals. Though 1-Way ANOVA examples we did in class like the paper airplane study had fewer than this, with more than 20 data points, our assumption tests could be more robust and more accurate of the true population when answering our hypothesis or SRQ. A larger data set in our case would be even more stadiums sampled, which could contain more variations in field size dimensions, and more examples of the four field categories (small, medium, large, XL). Other than that, there aren't many errors in the data values itself because it comes from official league statistics for the 2023 – 2024 season.

References

The data we used is the StadiumGoals_dataset.xlsx. This is an excel sheet with 20 row entries.

Others can access the data and view it here: https://pennstateoffice365-my.sharepoint.com/:x/g/personal/sik5675_psu_edu/EYTd-Y-oLndGqdAuKO0gVuAB_HS6uQDndlvLgfoa54fqIw?e=zbv7DZ

Code Appendix

```
# loading and prepping data
```

```
library(openxlsx)
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
soccerData <- read.xlsx(
```

```
  xlsxFile = "StadiumGoals_dataset.xlsx")
```

```
# remove stadium, team, size, and games columns
```

```
soccerData <- soccerData[, -c(1, 2, 3, 6)]
```

```
soccerData$`Goals.(season-23/24)` <- soccerData$`Goals.(season-23/24)` / 17
```

```
names(soccerData)[names(soccerData) == "Size.Cateogory"] <- "Size.Category"
```

```
soccerData$Size.Category <- as.factor(soccerData$Size.Category)
```

```
# fit model
```

```
soccerModel <- aov(
```

```
  formula = `Goals.(season-23/24)` ~ Size.Category,
```

```
  data = soccerData
```

```
)
```

```
# relationship between block and response
```

```
ggplot(
```

```
  data = soccerData,
```

```
  mapping = aes(
```

```
    y = `Goals.(season-23/24)`,
```

```
x = League
)
)+
geom_point(size = 2) +
geom_smooth(
  method = "lm",
  formula = y ~ x,
  color = "black",
  linetype = "dashed",
  se = FALSE
) +
theme_bw() +
labs(
  x = "League",
  y = "Goals",
  title = 'Relationship Between League and Goals Scored'
)
```

```
# residual plot
car::qqPlot(
  x = residuals(soccerModel),
  distribution = "norm",
  envelope = 0.90,
  id = FALSE,
  pch = 20,
  ylab = "Residuals"
)
```

```

# strip chart
ggplot(
  data = data.frame(
    residuals = soccerModel$residuals,
    fitted = soccerModel$fitted.values
  ),
  mapping = aes(x = fitted, y = residuals)
) +
  geom_point(size = 2) +
  theme_bw() +
  labs(
    x = "Fitted values (points)",
    y = "Residuals (points)",
    title = "Strip Chart"
  )

# independence of observations
ggplot(
  data = data.frame(
    residuals = soccerModel$residuals,
    index = 1:length(soccerModel$residuals)
  ),
  mapping = aes(x = index, y = residuals)
) +
  geom_point(size = 1.5) +
  geom_line() +

```

```

theme_bw() +
geom_hline(
  yintercept = 0,
  linetype = "dashed",
  color = "red"
) +
labs(
  x = "Measurement order",
  y = "Residuals",
  title = "Index Plot"
)
car::durbinWatsonTest(soccerModel)$dw

```

```

# advanced anova table
parameters::model_parameters(
  model = soccerModel,
  es_type = c("eta", "omega", "epsilon")
) %>%
knitr::kable(
  digits = 4,
  col.names = c(
    "Source", "SS", "df", "MS", "F", "p-value",
    "Eta Sq.", "Omega Sq.", "Epsilon Sq."),
  booktabs = TRUE,
  align = c("l", rep("c", 8))
) %>%

```

```
kableExtra::kable_styling(  
  font_size = 12,  
  latex_options = c("HOLD_position")  
)
```